LOOKING GLASS — DESIGN BRIEF

Visual System: Nothing OS × Liquid Glass WebGPU

Version: 2.0 — Consolidated
Sources: `dominikmartn/nothing-design-skill` v3.0.0 · `jeantimex/glass-effect-webgpu`

PART I — PHILOSOPHY & DESIGN DNA

Core Thesis

Nothing's design language is the skeleton — monochrome, typographic, brutally honest about structure. The WebGPU liquid glass is the skin — live refraction, physical depth, spring-based deformation. Neither works alone. Together they create a UI that reads like precision hardware and feels like touching water.

Glass is not decoration. Glass IS structure made visible.

When a toolbar floats above the canvas, the glass proves it's above — not via shadow or z-index tricks, but via actual refraction of what's behind it. Nothing's philosophy of honesty meets glass's physics of honesty. They are the same principle expressed differently.

Three Laws That Cannot Be Broken

1. **Glass reveals, it never obscures.** Every glassed element must allow the content behind it to remain readable through the distortion. If glass kills legibility, remove the glass.
2. **Monochrome is the canvas. Glass is the event.** The Nothing color system remains inviolate. Glass does not introduce color. It bends and reflects the existing monochrome world.
3. **Typography is sacred.** Text sits above glass on its own layer, never refracted. The hierarchy `display → body → metadata` is the only hierarchy that matters for reading.

Nothing Design Principles (from SKILL.md v3.0.0)

- **Subtract, don't add.** Every element must earn its pixel. Default to removal.
- **Structure is ornament.** Expose the grid, the data, the hierarchy itself.

- **Type does the heavy lifting.** Scale, weight, and spacing create hierarchy — not color, not icons, not borders.
- **Both modes are first-class.** Dark mode: OLED black. Light mode: warm off-white. Neither is derived from the other — both get full design attention.
- **Industrial warmth.** Technical and precise, but never cold.
- **Percussive, not fluid.** Transitions feel like clicks, not swooshes.

Anti-Patterns — Never Do These

Anti-pattern	How Looking Glass respects it
No gradients in UI chrome	All surfaces are flat or glass. Zero gradient fills.
No shadows outside glass	Card depth = glass shadow only. No additional `box-shadow`.
No skeleton loading screens	Cards show `[LOADING...]` in Space Mono during fetch.
No toast popups	Status shown inline: `[SAVED]` at card level, `[SYNCED]` in toolbar.
No mascots or empty state illustrations	Empty canvas: Doto `0` + one line of Space Mono copy. Nothing else.
No zebra striping	Context menu items: spacing alone distinguishes rows.
No filled icons	Phosphor Regular outline weight only.
No parallax or scroll-jacking	Canvas pan is 1:1 with pointer. No momentum, no inertia.
No spring/bounce in UI chrome	Spring only for physical glass deformation and toggle knob.
No border-radius > 16px on cards	Cards: 12px. Palette: 16px. Bottom sheet top corners: 20px maximum.

PART II — DUAL MODE COLOR SYSTEM

Both modes are first-class. Dark mode is OLED black. Light mode is warm off-white. Neither is an inversion of the other — they are two distinct materials.

Dark mode: contrast comes from luminance difference (white text on near-black surface). Glass glows — the specular highlight is the defining edge against the void.

Light mode: contrast comes from weight and proximity (dark text on warm paper, separated by space). Glass casts — the shadow is the defining edge that proves physical depth.

CSS Custom Properties

```
/* ── DARK MODE (default) ─────────────────────────── */
:root,
[data-theme="dark"] {
  /* Surfaces */
  --color-bg:             #0A0A0A;   /* OLED black — never pure #000 */
  --color-bg-raised:      #111111;
  --color-bg-overlay:     #181818;
  --color-bg-card:        #141414;

  /* Borders */
  --color-border:         rgba(255, 255, 255, 0.08);
  --color-border-active:  rgba(255, 255, 255, 0.20);
  --color-border-focus:   rgba(255, 255, 255, 0.35);

  /* Gray scale */
  --color-gray-900:       #1A1A1A;
  --color-gray-800:       #2A2A2A;
  --color-gray-600:       #555555;
  --color-gray-400:       #888888;
  --color-gray-200:       #BBBBBB;

  /* Text — four stops only */
  --text-display:         #FFFFFF;   /* 100% — hero numbers, one per screen */
  --text-primary:         #F5F5F5;   /* 90%  — body, card titles */
  --text-secondary:       #999999;   /* 60%  — metadata, labels */
  --text-disabled:        #444444;   /* 40%  — disabled, timestamps */

  /* Accent — an interrupt, never a decoration */
  --color-accent:         #D71921;   /* Nothing red. Only for urgency. */
  --color-success:        #22C55E;
  --color-warning:        #F59E0B;

  /* Canvas */
  --canvas-dot-color:     rgba(255, 255, 255, 0.10);
  --canvas-grid-color:    rgba(255, 255, 255, 0.04);
  --canvas-bg:            #0A0A0A;
```

```css
  /* Glass */
  --glass-tint:           rgba(255, 255, 255, 0.04);
  --glass-specular:       rgba(255, 255, 255, 0.25);
  --glass-bezel-shadow:   rgba(0,   0,   0,   0.12);
  --glass-cast-shadow:    rgba(0,   0,   0,   0.55);
  --glass-frost:          rgba(10,  10,  10,  0.60);

  /* States */
  --state-hover:          rgba(255, 255, 255, 0.06);
  --state-active:         rgba(255, 255, 255, 0.10);
  --state-focus-ring:     2px solid rgba(255, 255, 255, 0.60);
}

/* ── LIGHT MODE ──────────────────────────────────── */
[data-theme="light"] {
  /* Surfaces — warm off-white paper, never clinical white */
  --color-bg:             #F5F2EE;
  --color-bg-raised:      #EDE9E3;
  --color-bg-overlay:     #E5E0D8;
  --color-bg-card:        #F8F6F2;

  /* Borders */
  --color-border:         rgba(0, 0, 0, 0.08);
  --color-border-active:  rgba(0, 0, 0, 0.18);
  --color-border-focus:   rgba(0, 0, 0, 0.32);

  /* Gray scale — warm toned */
  --color-gray-900:       #E0DBD3;
  --color-gray-800:       #C8C2B8;
  --color-gray-600:       #8C877F;
  --color-gray-400:       #5A5550;
  --color-gray-200:       #2E2B27;

  /* Text */
  --text-display:         #0A0A0A;
  --text-primary:         #1A1815;
  --text-secondary:       #6B6560;
  --text-disabled:        #B0AA9F;

  /* Accent — unchanged */
  --color-accent:         #D71921;
  --color-success:        #16A34A;
  --color-warning:        #D97706;

  /* Canvas */
  --canvas-dot-color:     rgba(0, 0, 0, 0.08);
  --canvas-grid-color:    rgba(0, 0, 0, 0.04);
```

```
  --canvas-bg:            #F5F2EE;

  /* Glass */
  --glass-tint:           rgba(255, 255, 255, 0.55);
  --glass-specular:       rgba(255, 255, 255, 0.70);
  --glass-bezel-shadow:   rgba(0,   0,   0,   0.08);
  --glass-cast-shadow:    rgba(0,   0,   0,   0.15);
  --glass-frost:          rgba(240, 236, 229, 0.75);

  /* States */
  --state-hover:          rgba(0, 0, 0, 0.05);
  --state-active:         rgba(0, 0, 0, 0.10);
  --state-focus-ring:     2px solid rgba(0, 0, 0, 0.50);
}
```

Color Rules

- `--text-display` is used once per screen — the hero element only.
- `--color-accent` (#D71921) is an interrupt signal. If nothing is urgent, it does not appear.
- Status colors (`--color-success` , `--color-warning`) are applied to data values only, never to labels or row backgrounds.
- Glass does not introduce color. It bends the existing monochrome palette.
- Text of any size or weight is never refracted or glassed.

PART III — TYPOGRAPHY

Font Stack

```
@import url('https://fonts.googleapis.com/css2?family=Space+Grotesk:wght@300;400;

--font-display: 'Doto', 'Space Grotesk', monospace;
--font-body:    'Space Grotesk', system-ui, sans-serif;
--font-ui:      'Space Mono', 'Courier New', monospace;
```

Three-Layer Hierarchy — Strict

Every screen has exactly three layers of importance. Not two, not five. Three.

Layer	Font	Size	Weight	Case	Use
DISPLAY	Doto or Space Grotesk	48–96px	300–700	Any	One hero element per screen only. Item counts, canvas name.
BODY	Space Grotesk	13–22px	400–600	Sentence	Card titles, note text, descriptions, modal content.
METADATA	Space Mono	10–13px	400	ALL CAPS	Timestamps, domains, tags, labels, toolbar captions.

Type Scale

```
--text-hero:    72px;   /* Doto/Grotesk hero — one per view maximum */
--text-display: 48px;   /* Display numbers */
--text-2xl:     28px;   /* Large heading */
--text-xl:      22px;   /* Subheading */
--text-lg:      18px;   /* Card title large */
--text-base:    15px;   /* Body text */
--text-sm:      13px;   /* Secondary body */
--text-xs:      11px;   /* Metadata, labels */
--text-2xs:     9px;    /* Tiny labels, toolbar captions */
```

Font Discipline

Per screen use maximum: 2 font families, 3 font sizes, 2 font weights. Every additional size or weight costs visual coherence.

Typography Rules for Glass Contexts

- Add `text-shadow: 0 1px 2px rgba(0,0,0,0.6)` to all body text above glass surfaces.
- Never set opacity < 0.9 for BODY or DISPLAY text above glass.
- Space Mono metadata uses `letter-spacing: 0.12em` above glass.
- The text-shadow value above applies only to dark mode. In light mode it is not needed — contrast is sufficient.

PART IV — GLASS EFFECT SPECIFICATION

How Glass Differs Between Modes

Dark mode glass is luminous. The specular highlight is the defining edge — it separates the glass surface from the dark void behind it. `specularIntensity` is the primary parameter.

Light mode glass is physical. The cast shadow is the defining edge — it proves the glass is above the warm paper surface. `shadowIntensity` is the primary parameter.

WebGPU Uniforms — Dark Mode

```
const GLASS_MODE_DARK = {
  specularIntensity:    0.30,
  shadowIntensity:      0.15,
  backgroundBrightness: 1.08,
  tintStrength:         0.04,
}
```

WebGPU Uniforms — Light Mode

```
const GLASS_MODE_LIGHT = {
  specularIntensity:    0.55,
  shadowIntensity:      0.40,
  backgroundBrightness: 0.96,
  tintStrength:         0.45,
}
```

Five Glass Surfaces

The app has exactly five distinct glass surfaces. Each has different refraction parameters reflecting physical distance from content.

GLASS-1 — Canvas Cards

Cards feel like frosted acrylic resting on the canvas, not lenses distorting what's behind them.

```
const CARD_GLASS_DARK: GlassUniforms = {
  refractionStrength: 0.08,
  blurRadius: 6.0,
  specularIntensity: 0.20,
  shadowIntensity: 0.10,
```

```
  thickness: 0.04,
  bezels: { tl: 12, tr: 12, br: 12, bl: 12 },
  liquidDeformation: true,
  deformationRadius: 0.3,
  springStiffness: 300,
  springDamping: 20,
}

const CARD_GLASS_LIGHT: GlassUniforms = {
  ...CARD_GLASS_DARK,
  refractionStrength: 0.05,
  specularIntensity: 0.40,
  shadowIntensity: 0.25,
}
```

Card anatomy:

```
┌──────────────────────────────────┐  ← Specular arc (1px, 25% white)
│  [OG IMAGE / MEDIA]              │  ← Full-bleed, no glass over image
├──────────────────────────────────┤  ← 1px separator: rgba(255,255,255,0.06) dark
│  CARD TITLE                      │  ← Space Grotesk 15px, --text-primary
│  Short description text here     │  ← Space Grotesk 13px, --text-secondary
│                                  │
│  DOMAIN.COM          2H AGO      │  ← Space Mono 10px ALL CAPS
└──────────────────────────────────┘  ← Bezel shadow (1px, 12% black)
```

Card accent dot: 4×4px in top-right corner, one of five status colors. The only colored element. Sits above the glass layer, never refracted.

GLASS-2 — Toolbar

The toolbar feels like a premium hardware control surface — physically distinct from the canvas but not competing with content.

```
const TOOLBAR_GLASS_DARK: GlassUniforms = {
  refractionStrength: 0.22,
  blurRadius: 20.0,
  specularIntensity: 0.30,
  shadowIntensity: 0.40,
  thickness: 0.10,
  bezels: { tl: 20, tr: 20, br: 20, bl: 20 },
  liquidDeformation: false,
}
```

```
const TOOLBAR_GLASS_LIGHT: GlassUniforms = {
  ...TOOLBAR_GLASS_DARK,
  refractionStrength: 0.18,
  specularIntensity: 0.50,
  shadowIntensity: 0.30,
}
```

GLASS-3 — Command Palette

The command palette is the hero glass element — a large sharp pane catching ambient light above everything.

```
const COMMAND_GLASS_DARK: GlassUniforms = {
  refractionStrength: 0.35,
  blurRadius: 32.0,
  specularIntensity: 0.45,
  shadowIntensity: 0.60,
  thickness: 0.16,
  bezels: { tl: 16, tr: 16, br: 16, bl: 16 },
  liquidDeformation: false,
  backdropDim: 0.30,
}

const COMMAND_GLASS_LIGHT: GlassUniforms = {
  ...COMMAND_GLASS_DARK,
  refractionStrength: 0.28,
  specularIntensity: 0.60,
  shadowIntensity: 0.25,
}
```

GLASS-4 — Context Menu

Small and surgical. No liquid deformation.

```
const CONTEXT_GLASS_DARK: GlassUniforms = {
  refractionStrength: 0.15,
  blurRadius: 14.0,
  specularIntensity: 0.20,
  shadowIntensity: 0.50,
  thickness: 0.08,
```

```
  bezels: { tl: 8, tr: 8, br: 8, bl: 8 },
  liquidDeformation: false,
}

const CONTEXT_GLASS_LIGHT: GlassUniforms = {
  ...CONTEXT_GLASS_DARK,
  refractionStrength: 0.12,
  specularIntensity: 0.35,
  shadowIntensity: 0.20,
}
```

GLASS-5 — Lightbox / Media Viewer

Controls overlay uses the thickest glass. Cinematic — content is the star, glass is the stage.

```
const LIGHTBOX_GLASS_DARK: GlassUniforms = {
  refractionStrength: 0.40,
  blurRadius: 40.0,
  specularIntensity: 0.50,
  shadowIntensity: 0.70,
  thickness: 0.20,
  bezels: { tl: 24, tr: 24, br: 24, bl: 24 },
  liquidDeformation: true,
  deformationRadius: 0.5,
  springStiffness: 400,
  springDamping: 25,
}

const LIGHTBOX_GLASS_LIGHT: GlassUniforms = {
  ...LIGHTBOX_GLASS_DARK,
  refractionStrength: 0.32,
  specularIntensity: 0.65,
  shadowIntensity: 0.30,
}
```

Glass Refraction Summary

Surface	Dark strength	Light strength
Cards	0.08	0.05
Toolbar	0.22	0.18

Command palette	0.35	0.28
Context menu	0.15	0.12
Lightbox controls	0.40	0.32

WebGPU Fallback — Three Tiers

```
// src/utils/glassTier.ts
export const getGlassTier = async (): Promise<'webgpu' | 'css-backdrop' | 'flat'>
  if (navigator.gpu) {
    const adapter = await navigator.gpu.requestAdapter()
    if (adapter) return 'webgpu'
  }
  if (CSS.supports('backdrop-filter', 'blur(1px)') ||
      CSS.supports('-webkit-backdrop-filter', 'blur(1px)')) {
    return 'css-backdrop'
  }
  return 'flat'
}
```

Tier 2 — CSS backdrop-filter (Safari, Firefox):

```
[data-theme="dark"] .glass-surface {
  background: rgba(14, 14, 14, 0.70);
  backdrop-filter: blur(20px) saturate(110%);
  -webkit-backdrop-filter: blur(20px) saturate(110%);
  border: 1px solid rgba(255, 255, 255, 0.08);
  box-shadow:
    inset 0 1px 0 rgba(255, 255, 255, 0.20),
    0 8px 32px rgba(0, 0, 0, 0.55);
}

[data-theme="light"] .glass-surface {
  background: rgba(240, 236, 229, 0.75);
  backdrop-filter: blur(20px) saturate(130%);
  -webkit-backdrop-filter: blur(20px) saturate(130%);
  border: 1px solid rgba(0, 0, 0, 0.08);
  box-shadow:
    inset 0 1px 0 rgba(255, 255, 255, 0.65),
    0 4px 16px rgba(0, 0, 0, 0.15),
    0 1px 4px rgba(0, 0, 0, 0.10);
}
```

Tier 3 — Flat (legacy / low-power mode):

```
[data-theme="dark"] .glass-surface {
  background: #1A1A1A;
  border: 1px solid rgba(255, 255, 255, 0.10);
}

[data-theme="light"] .glass-surface {
  background: #EDE9E3;
  border: 1px solid rgba(0, 0, 0, 0.10);
}
```

PART V — SPACING SYSTEM

Spacing is the primary layout tool. Proximity creates grouping. Distance creates separation. If a divider line is needed, the spacing is wrong.

```
--space-1:   4px;    /* Icon + label coupling */
--space-2:   8px;    /* Same element, different lines */
--space-3:   12px;   /* Related items within a group */
--space-4:   16px;   /* Items in the same section */
--space-5:   20px;   /* Card internal padding */
--space-6:   24px;   /* Section breathing room */
--space-8:   32px;   /* Between groups */
--space-10:  40px;   /* Major section separation */
--space-12:  48px;   /* Hero to content transition */
--space-16:  64px;   /* New context begins */
--space-24:  96px;   /* Vast — hero moments only */
```

Applied Spacing

```
CARD INTERNAL:
  Image area: 0 padding (full-bleed)
  Separator:  1px rgba(255,255,255,0.06) dark / rgba(0,0,0,0.06) light
  Content horizontal padding: 16px
  Content vertical padding: 12px
  Title to description: 6px
  Description to metadata row: 12px
  Metadata row: 16px horizontal, flush to card bottom edges

TOOLBAR:
  Icon to icon: 8px
  Icon to label: 4px
  Group to group: 24px
```

```
  Edge padding: 16px

COMMAND PALETTE:
  Input vertical padding: 20px
  Result item height: 48px
  Section header to first result: 8px
  Outer padding: 24px

CONTEXT MENU:
  Item height: 40px
  Item horizontal padding: 16px
  Divider margin: 4px each side
```

The Divider Exception

Dividers are permitted in exactly two places in the entire app:

1. Inside a card — the separator between image and text content
2. Context menu — between action groups

Nowhere else. All other surfaces use spacing alone.

PART VI — ICONOGRAPHY

```
Source:   Phosphor Icons (outline weight only)
          pnpm add @phosphor-icons/react

Sizes:    16px — toolbar, metadata, inline
          20px — card actions, context menu
          24px — empty state, primary action spots

Weight:   Regular outline. No Bold, no Fill, no Duotone.

Color:    --text-secondary at rest
          --text-primary on hover/active
          --color-accent on destructive actions only

Hitbox:   32×32px minimum desktop
          44×44px minimum touch (iOS HIG)
```

Icon Vocabulary

Action	Phosphor Name
Add item	`Plus`
Add URL	`ArrowUpRight`
Search / Command	`Command`
Spaces	`SquaresFour`
Settings	`GearSix`
Close	`X`
Drag handle	`DotsSixVertical`
Group	`Brackets`
Archive	`Archive`
Delete	`Trash`
Bookmark type	`Bookmark`
Web clip type	`Globe`
Note type	`Note`
Image type	`Image`
Video type	`Play`
PDF type	`FilePdf`
Export	`Export`
Import	`Download`
Zoom fit	`CornersOut`
Minimap	`MapTrifold`
Undo	`ArrowCounterClockwise`
Redo	`ArrowClockwise`
Light mode	`Sun`

Dark mode	`Moon`
Link	`Link`
Copy	`Copy`
Edit	`PencilSimple`
Open external	`ArrowSquareOut`
Collapse	`CaretUp`
Expand	`CaretDown`
AI / Ask	`Sparkle`

Only icons from this vocabulary are used. If a function cannot be represented by this set, reconsider the function before adding an icon.

Icon + Label Pairing

Desktop toolbar icons have a label below them. Label: Space Mono 9px ALL CAPS, `--text-secondary`, 4px gap. On mobile, labels are hidden — icon alone carries meaning.

PART VII — COMPONENT SPECIFICATIONS

Cards — Five States

```
REST:     refractionStrength=0.08, shadow=0.10, scale 1.0
HOVER:    specularIntensity +0.08, shadow 0.20, overlay --state-hover
DRAG:     refractionStrength=0.20, shadow=0.55, scale(1.02), rotation ±2deg
          Glass deforms (liquidDeformation=true, spring fires)
SELECTED: border: 1px solid rgba(255,255,255,0.35) dark / rgba(0,0,0,0.25) light
FOCUS:    --state-focus-ring at 3px offset
DROP:     Spring settles to REST values over 280ms
          Brief ripple: liquidDeformation fires outward from drop point
```

Card Differences by Mode

Property	Dark	Light

Background	`#141414` + glass	`#F8F6F2` + glass
Separator	`rgba(255,255,255,0.06)`	`rgba(0,0,0,0.06)`
Title	`#F5F5F5`	`#1A1815`
Metadata text	`#999999`	`#6B6560`
Accent dot	Same 5 colors	Same 5 colors
Drag shadow	`rgba(0,0,0,0.60)`	`rgba(0,0,0,0.18)`

Buttons

Primary (glass surface):

```
Glass:      refractionStrength=0.15, blurRadius=8px
Label:      Space Mono 12px ALL CAPS, --text-display
Height:     40px, min-width: 120px, radius: 8px
Hover:      specularIntensity → 0.40, scale(1.01)
Press:      liquidDeformation fires inward 4px, spring settles 180ms
Focus:      --state-focus-ring
Disabled:   opacity 0.35
```

Secondary (border only, no glass):

```
Border:     1px solid --color-border
Label:      Space Mono 12px ALL CAPS, --text-secondary
Hover:      border → --color-border-active
```

Ghost / icon:

```
Icon only, 32×32px hitbox
Hover:      background --state-hover, 6px radius
Active:     background --state-active, scale(0.92)
```

Segmented Controls

```
Track:          1px border --color-border, no background fill
Active pill:    Glass (refractionStrength=0.12, blurRadius=6px)
                Slides within track on selection change
                Transition: 220ms cubic-bezier(0.34, 1.56, 0.64, 1)
Label:          Space Mono 11px ALL CAPS
Active label:   --text-display
```

```
Inactive label: --text-disabled
Height:         36px
```

Progress / Status Bars

```
20 equal segments, 2px gap between each
Filled:     rgba(255,255,255,0.85) dark / rgba(0,0,0,0.70) light
            3px tall, 2px radius
Empty:      rgba(255,255,255,0.12) dark / rgba(0,0,0,0.10) light
Animation:  fill from left, each segment delays 20ms from previous
No glass on progress bars — they are data, not chrome.
Label:      Space Mono 11px ALL CAPS — "ITEMS  ████░░░░  248 / 400"
```

Toggle Switch

```
Track:        28×16px, border 1px --color-border
Knob:         Glass circle (refractionStrength=0.20, preset: circle), 12×12px
Labels:       Space Mono 11px ALL CAPS flanking the track — no icons
              "DARK  ●────────  LIGHT"
On state:     Knob slides right, specular flips to right arc
Animation:    Spring stiffness 500, damping 30 (mechanical click feel)
Disabled:     opacity 0.35, spring disabled
```

The mode toggle uses this component. It is the only toggle that uses word labels instead of an icon.

URL Input Field

```
Glass:        refractionStrength=0.12, blurRadius=10px
Border:       1px --color-border at rest → --color-border-focus on focus
              No accent color on focus — Nothing discipline
Icon:         ArrowUpRight, 16px, --text-secondary, left side
Placeholder: Space Grotesk 14px, --text-disabled
Input text:  Space Grotesk 14px, --text-primary
Height:       48px, radius: 12px
Focus:        specularIntensity → 0.35, scale(1.005)
Error:        border → --color-accent
              Error copy: Space Mono 11px ALL CAPS, --color-accent
              Format: "[ERROR: URL UNREACHABLE]" not sentence-case prose
```

Context Menu

```
Structure:

  ╔════════════════════╗
  ║  OPEN              ║  ← Space Grotesk 14px, --text-primary
  ║  EDIT              ║
  ║  COPY LINK         ║
  ╠════════════════════╣  ← 1px divider
  ║  GROUP             ║
  ║  COLOR  ● ● ● ● ●  ║  ← 5 color dots, 4px circles
  ╠════════════════════╣
  ║  ARCHIVE           ║
  ║  DELETE            ║  ← --color-accent (only red item)
  ╚════════════════════╝

Width:         180px
Item height:   40px
H-padding:     16px
Glass:         CONTEXT_GLASS uniforms
```

Command Palette

```
Structure:

  ┌────────────────────────────────────────────┐
  │  ◉  Search or paste a URL...               │  ← Space Grotesk 16px placeholder
  ├────────────────────────────────────────────┤
  │  RECENT                                    │  ← Space Mono 10px ALL CAPS
  │  ↗  figma.com/file/...                    │  ← Space Grotesk 14px result
  │                                            │
  │  ACTIONS                                   │  ← Space Mono 10px ALL CAPS
  │  ⊞  New Note                              │
  │  ⊞  New Space                             │
  └────────────────────────────────────────────┘

Width:         min(640px, 90vw)
Glass:         COMMAND_GLASS uniforms
Keyboard:      ↑↓ navigate, Enter select, Esc close
```

Minimap

```
Size:          160×100px desktop, hidden mobile
Background:    rgba(0,0,0,0.60) — not glass
Item dots:     2px, --text-secondary opacity
Viewport:      Glass rectangle (refractionStrength=0.30, specularIntensity=0.40)
Corner:        Bottom-right, 8px radius
Interaction:   Click → pan canvas to location
```

```
               Drag viewport rect → live pan
Update rate:   30fps throttle
```

Empty States

```
EMPTY CANVAS:
  "0 ITEMS"                  ← Doto or Space Mono 48px, --text-disabled
  "PASTE A URL TO BEGIN"     ← Space Mono 13px ALL CAPS, --text-disabled, 24px belo

EMPTY SEARCH:
  "[NO RESULTS]"             ← Space Mono 13px ALL CAPS, --text-disabled, inline

LOADING:
  "[LOADING...]"             ← Space Mono 11px ALL CAPS, --text-disabled

ERROR:
  "[ERROR: FETCH FAILED]"    ← Space Mono 11px ALL CAPS, --color-accent
  "[RETRY]"                  ← Same line, --text-secondary, 8px gap
```

PART VIII — CANVAS BACKGROUND

```
/* Dark mode — white dots on OLED black */
[data-theme="dark"] #canvas-world {
  background-color: #0A0A0A;
  background-image: radial-gradient(circle, rgba(255,255,255,0.10) 1px, transpare
  background-size: 32px 32px;
}

/* Light mode — dark dots on warm paper */
[data-theme="light"] #canvas-world {
  background-color: #F5F2EE;
  background-image: radial-gradient(circle, rgba(0,0,0,0.08) 1px, transparent 1px
  background-size: 32px 32px;
}
```

The dot matrix is Nothing's signature pattern. Glass over dots shows beautiful refraction — dots bend and distort through the glass lens. This is the primary reason the dot grid is the default, not just aesthetic preference.

Alternative grids (user-selectable):

- Fine line grid: 1px rgba lines at 40px intervals, 4% opacity

- Plain surface: No grid, background color only

PART IX — MOTION SYSTEM

Core Principle

Spring physics applies to physical glass deformation only. All CSS transitions use ease-out. Nothing's "percussive, not fluid" rule means UI chrome moves decisively — it does not swoosh or bounce.

CSS Transitions

```
--ease-out:    cubic-bezier(0.0, 0.0, 0.2, 1);
--ease-in:     cubic-bezier(0.4, 0.0, 1.0, 1);
--ease:        cubic-bezier(0.4, 0.0, 0.2, 1);

--duration-instant:    80ms;    /* Hover feedback */
--duration-fast:       160ms;   /* Icon state, color change */
--duration-standard:   240ms;   /* Panel open/close */
--duration-deliberate: 320ms;   /* Canvas enter, lightbox open */
--duration-canvas:     200ms;   /* Space switch crossfade */
```

Spring Physics Reference

Surface	Stiffness	Damping	Feel
Card drag pickup	300	20	Elastic, snappy
Card drop settle	250	22	Gentle, resolved
Toolbar slide in	400	28	Crisp
Context menu appear	500	30	Instant, precise
Command palette open	200	18	Fluid, deliberate
Lightbox open	180	16	Cinematic
Toggle switch	500	30	Mechanical click
Button press deform	400	25	Physical press
Bottom sheet snap	400	30	Decisive snap

Bottom sheet dismiss	N/A	N/A	220ms ease-in (no spring)

What Animates vs What Doesn't

Element	Animates?	What
Card hover	✅	Glass specular, shadow
Card drag lift	✅	Scale, shadow, glass, rotation
Card drop	✅	Spring settle to rest
Card position change (non-drag)	❌	Instant
Toolbar icon hover	✅	Background opacity, 80ms
Toolbar appear	✅	translateY, 240ms ease-out
Command palette	✅	opacity + scale(0.98→1.0), 200ms
Context menu	✅	opacity + scale(0.96→1.0), 120ms
Bottom sheet	✅	Spring to snap point
Space switch	✅	Canvas opacity crossfade 200ms
Mode switch	✅	CSS vars 280ms ease-out
Glass deformation	✅	Spring physics (WebGPU)
Canvas pan	❌	1:1 with pointer, never animated
Canvas zoom	❌	Instant
Typography	❌	Never transitions
Layout changes	❌	Always instant

Mode Switch Behaviour

```
:root {
  transition: background-color 280ms ease-out, color 280ms ease-out;
}
/* The canvas world (#canvas-world) switches instantly — a crossfade during
   pan/zoom looks broken. Only chrome (toolbar, cards, palette) fades. */
```

WebGPU glass uniforms animate simultaneously on theme-change:

```
window.addEventListener('theme-change', (e: CustomEvent) => {
  const isDark = e.detail.theme === 'dark'
  Object.entries(glassSurfaces).forEach(([id, surface]) => {
    surface.updateUniforms({
      specularIntensity: isDark ? DARK_UNIFORMS[id].specularIntensity : LIGHT_UNI
      shadowIntensity:   isDark ? DARK_UNIFORMS[id].shadowIntensity   : LIGHT_UNI
      tintStrength:      isDark ? DARK_UNIFORMS[id].tintStrength       : LIGHT_UN
      transitionDuration: 320,
    })
  })
})
```

PART X — STATES SYSTEM

Every interactive element has exactly five states. None may be omitted.

```
REST         → Default. Lowest visual weight.
HOVER        → Pointer over element. Subtle signal.
ACTIVE/PRESS → Pointer/touch down. Physical confirmation.
FOCUS        → Keyboard focus. Visible without color alone.
DISABLED     → Not interactive. 35% opacity, default cursor.
```

State Overlays

```
/* Applied on top of base styles per mode */
[data-theme="dark"] {
  --state-hover:  rgba(255, 255, 255, 0.06);
  --state-active: rgba(255, 255, 255, 0.10);
  --state-focus-ring: 2px solid rgba(255, 255, 255, 0.60);
  --state-focus-offset: 3px;
  --state-disabled-opacity: 0.35;
}
[data-theme="light"] {
  --state-hover:  rgba(0, 0, 0, 0.05);
  --state-active: rgba(0, 0, 0, 0.10);
  --state-focus-ring: 2px solid rgba(0, 0, 0, 0.50);
  --state-focus-offset: 3px;
  --state-disabled-opacity: 0.35;
}
```

PART XI — SPACES (MULTI-CANVAS SYSTEM)

Each Space is a named independent infinite canvas. The Spaces UI feels like a hardware panel switcher.

Desktop Sidebar

```
Width: 240px overlay (slides over canvas, does not push it)
Glass: TOOLBAR_GLASS uniforms

┌──────────────────────────────────────┐
│  SPACES                              │  ← Space Mono 10px ALL CAPS
│                                      │
│  ● MAIN                      14      │  ← Active: Space Grotesk 14px 600wt
│  ○ RESEARCH                  8       │  ← Inactive: Space Grotesk 14px 400wt
│  ○ MOODBOARD                 23      │
│  ○ ARCHIVE                   156     │
│  ─────────────────────────────────   │
│  + NEW SPACE                         │  ← Space Mono 12px ALL CAPS
└──────────────────────────────────────┘

Active indicator:   3×3px filled dot (●), --text-primary
Inactive indicator: 3×3px outline circle (○), --text-disabled
Count:              Space Mono 11px, right-aligned, --text-secondary
Hover:              Background --state-hover, name → --text-primary
Drag handle:        DotsSixVertical 12px, appears on hover only
```

Space Switching Transition

```
1. Current canvas fades to 0% opacity: 80ms ease-in
2. New canvas fades to 100% opacity: 120ms ease-out
3. Toolbar and minimap do NOT fade — persistent chrome
4. Viewport state (pan position, zoom) preserved per space
```

Mobile Spaces Bar

```
╔════════════════════════════════════════════════╗
║  ● MAIN    ○ RESEARCH    ○ MOODBOARD    ···    ║
╚════════════════════════════════════════════════╝
```

```
Horizontal scroll when > 3 spaces
Height: 44px + env(safe-area-inset-bottom)
Glass: TOOLBAR_GLASS
Active: Space Mono 11px ALL CAPS, --text-primary
Inactive: Space Mono 11px ALL CAPS, --text-disabled
"···" → bottom sheet with full spaces list
```

PART XII — MOBILE SPECIFICATIONS

Touch Targets

```
Minimum:              44×44px (iOS HIG absolute minimum)
Preferred:            48×48px for primary actions
Dangerous (delete):   44×44px with 8px separation from adjacent targets
Card drag zone:       Full card header (48px height)
```

Gesture System

```
CANVAS:
  One finger empty canvas:  Pan
  Two finger pinch:         Zoom
  Double tap empty:         Zoom to fit
  Long press empty:         Add item at that position

CARDS:
  One finger card header:   Drag
  Long press card body:     Context menu (bottom sheet)
  Double tap card:          Open / expand
  Pinch on selected card:   Resize
  Swipe left:               Quick archive
```

Bottom Sheet

```
Snap points:   15vh (peek) · 50vh (default) · 90vh (full)
Glass:         refractionStrength=0.28, blurRadius=28px, specularIntensity=0.35
Corners:       Top only: 20px radius. Bottom: 0 (flush to screen).
Handle:        36×4px, rgba(255,255,255,0.25) dark, centered, 10px from top
Backdrop:      rgba(0,0,0,0.40) dim, NOT glass
Dismiss:       Drag below peek point · Tap backdrop · ESC
Open spring:   stiffness 300, damping 28
Snap spring:   stiffness 400, damping 30
```

```
Close:         translateY off-screen, 220ms ease-in (no spring on close)
```

Mobile Toolbar

```
╔══════════════════════════════════════════════════════════╗
║  [⊞] MAIN      [+]        [⌘K]        [≡]        [...]  ║
╚══════════════════════════════════════════════════════════╝
Height: 56px + env(safe-area-inset-bottom)
Glass: TOOLBAR_GLASS uniforms
Labels: Space Mono 9px ALL CAPS, hidden if < 375px viewport width
```

iOS PWA Requirements

```
<meta name="apple-mobile-web-app-capable" content="yes">
<meta name="apple-mobile-web-app-status-bar-style" content="black-translucent">
<meta name="apple-mobile-web-app-title" content="Looking Glass">
<meta name="viewport" content="width=device-width, initial-scale=1, viewport-fit=
```

```
.toolbar-mobile {
  padding-bottom: calc(16px + env(safe-area-inset-bottom));
}
```

PART XIII — RESPONSIVE LAYOUT

Three genuinely distinct layout modes. Each is first-class.

```
MOBILE    0 – 767px:   Full screen canvas, bottom toolbar, no persistent rail
TABLET    768 – 1023px: 48px icon-only left rail, sidebar as overlay
DESKTOP   1024px+:     64px icon+label left rail, 240px expandable spaces panel
```

	Mobile	Tablet	Desktop
Toolbar	Bottom bar, 56px	Left rail, 48px	Left rail, 64px
Sidebar	Bottom sheet only	Overlay on tap	Persistent expandable
Card default width	min(90vw, 320px)	280px	300px
Canvas zoom default	0.75×	1.0×	1.0×

Minimap	Hidden	Optional	On by default

Component adaptation:

- Command palette: centered overlay (desktop/tablet) → full-width bottom sheet (mobile)
- Context menu: cursor-positioned (desktop/tablet) → bottom sheet (mobile)
- Toolbar labels: visible (desktop) → hidden (tablet) → hidden (mobile)

PART XIV — ACCESSIBILITY

Contrast Ratios

Pairing	Dark Mode	Light Mode	WCAG
`--text-primary` on `--color-bg`	19.4:1	16.8:1	AAA ✅
`--text-secondary` on `--color-bg`	8.2:1	4.8:1	AA ✅
`--color-accent` on `--color-bg`	4.6:1	4.1:1	AA ✅
`--text-disabled` on `--color-bg`	2.0:1	1.7:1	Exempt (disabled state)

Text above glass must pass contrast against the most transparent glass state, not the average.

Keyboard Navigation

```
Tab / Shift+Tab:         Focus next / previous card
Arrow keys:              Move focused card (8px per press, 40px + Shift)
Enter / Space:           Open focused card
Delete / Backspace:      Delete focused card
Cmd+Z / Ctrl+Z:          Undo
Cmd+Shift+Z / Ctrl+Y:    Redo
Cmd+K / Ctrl+K:          Open command palette
↑↓ in palette:           Navigate results
Escape:                  Close overlay / cancel / deselect
```

ARIA Roles

```
<div
  role="application"
  aria-label="Looking Glass canvas — MAIN space, 14 items"
  aria-describedby="canvas-instructions">

<p id="canvas-instructions" class="sr-only">
  Use Tab to navigate between items. Arrow keys move selected items.
  Press Enter to open. Press Delete to remove.
</p>

<article
  role="article"
  aria-label="[Card title] — [type] — added [date]"
  aria-posinset="[index]"
  aria-setsize="[total]"
  tabindex="0">

<nav aria-label="Looking Glass toolbar">
  <button aria-label="Spaces — currently viewing MAIN">
  <button aria-label="Add item">
  <button aria-label="Search and commands">

<div role="status" aria-live="polite" aria-atomic="true" class="sr-only" id="canv
<div role="alert" aria-live="assertive" class="sr-only" id="canvas-error">
```

Reduced Motion

```
@media (prefers-reduced-motion: reduce) {
  *, *::before, *::after {
    animation-duration:        0.01ms !important;
    animation-iteration-count: 1      !important;
    transition-duration:       0.01ms !important;
  }
}
```

WebGPU renderer checks `window.matchMedia('(prefers-reduced-motion: reduce)').matches` on init and disables `liquidDeformation` on all surfaces when true.

PART XV — COMPLETE TOKEN FILE

This is the single source of truth. Write this file to `src/styles/tokens.css`.

```
/* ══════════════════════════════════════════════════
   LOOKING GLASS — DESIGN TOKENS v2.0
   Nothing Design Skill v3.0.0 × WebGPU Glass
   ══════════════════════════════════════════════════ */

@import url('https://fonts.googleapis.com/css2?family=Space+Grotesk:wght@300;400;

:root {
  /* Fonts */
  --font-display: 'Doto', 'Space Grotesk', monospace;
  --font-body:    'Space Grotesk', system-ui, sans-serif;
  --font-ui:      'Space Mono', 'Courier New', monospace;

  /* Type scale */
  --text-hero:    72px;
  --text-display: 48px;
  --text-2xl:     28px;
  --text-xl:      22px;
  --text-lg:      18px;
  --text-base:    15px;
  --text-sm:      13px;
  --text-xs:      11px;
  --text-2xs:     9px;

  /* Weights */
  --weight-light:    300;
  --weight-regular:  400;
  --weight-medium:   500;
  --weight-semibold: 600;
  --weight-bold:     700;

  /* Line height */
  --leading-tight:  1.2;
  --leading-snug:   1.35;
  --leading-normal: 1.5;

  /* Letter spacing */
  --tracking-tight:   -0.02em;
  --tracking-normal:   0em;
  --tracking-wide:     0.06em;
  --tracking-widest:   0.15em;

  /* Spacing */
  --space-1:   4px;
```

```css
--space-2:   8px;
--space-3:   12px;
--space-4:   16px;
--space-5:   20px;
--space-6:   24px;
--space-8:   32px;
--space-10:  40px;
--space-12:  48px;
--space-16:  64px;
--space-24:  96px;

/* Radius */
--radius-xs:   2px;
--radius-sm:   4px;
--radius-md:   8px;
--radius-lg:   12px;
--radius-xl:   16px;
--radius-2xl:  20px;
--radius-pill: 9999px;

/* Motion */
--ease-out:    cubic-bezier(0.0, 0.0, 0.2, 1);
--ease-in:     cubic-bezier(0.4, 0.0, 1.0, 1);
--ease:        cubic-bezier(0.4, 0.0, 0.2, 1);
--duration-instant:    80ms;
--duration-fast:       160ms;
--duration-standard:   240ms;
--duration-deliberate: 320ms;
--duration-canvas:     200ms;

/* Z-index */
--z-canvas:       0;
--z-canvas-ui:    10;
--z-toolbar:      100;
--z-tooltip:      200;
--z-dropdown:     300;
--z-modal:        400;
--z-bottom-sheet: 500;
--z-lightbox:     600;
--z-notification: 700;
--z-debug:        9999;

/* Glass blur levels */
--glass-blur-sm:  6px;
--glass-blur-md:  14px;
--glass-blur-lg:  20px;
--glass-blur-xl:  32px;
```

```css
  --glass-blur-2xl: 40px;
}

/* ── DARK MODE ─────────────────────────────────── */
:root,
[data-theme="dark"] {
  --color-bg:            #0A0A0A;
  --color-bg-raised:     #111111;
  --color-bg-overlay:    #181818;
  --color-bg-card:       #141414;
  --color-border:        rgba(255, 255, 255, 0.08);
  --color-border-active: rgba(255, 255, 255, 0.20);
  --color-border-focus:  rgba(255, 255, 255, 0.35);
  --color-gray-900:      #1A1A1A;
  --color-gray-800:      #2A2A2A;
  --color-gray-600:      #555555;
  --color-gray-400:      #888888;
  --color-gray-200:      #BBBBBB;
  --text-display:        #FFFFFF;
  --text-primary:        #F5F5F5;
  --text-secondary:      #999999;
  --text-disabled:       #444444;
  --color-accent:        #D71921;
  --color-success:       #22C55E;
  --color-warning:       #F59E0B;
  --canvas-dot-color:    rgba(255, 255, 255, 0.10);
  --canvas-grid-color:   rgba(255, 255, 255, 0.04);
  --canvas-bg:           #0A0A0A;
  --glass-tint:          rgba(255, 255, 255, 0.04);
  --glass-specular:      rgba(255, 255, 255, 0.25);
  --glass-bezel-shadow:  rgba(0, 0, 0, 0.12);
  --glass-cast-shadow:   rgba(0, 0, 0, 0.55);
  --glass-frost:         rgba(10, 10, 10, 0.60);
  --state-hover:         rgba(255, 255, 255, 0.06);
  --state-active:        rgba(255, 255, 255, 0.10);
  --state-focus-ring:    2px solid rgba(255, 255, 255, 0.60);
  --state-focus-offset:  3px;
  --state-disabled-opacity: 0.35;
  transition: background-color 280ms var(--ease-out), color 280ms var(--ease-out)
}

/* ── LIGHT MODE ────────────────────────────────── */
[data-theme="light"] {
  --color-bg:            #F5F2EE;
  --color-bg-raised:     #EDE9E3;
  --color-bg-overlay:    #E5E0D8;
  --color-bg-card:       #F8F6F2;
```

```
  --color-border:        rgba(0, 0, 0, 0.08);
  --color-border-active: rgba(0, 0, 0, 0.18);
  --color-border-focus:  rgba(0, 0, 0, 0.32);
  --color-gray-900:      #E0DBD3;
  --color-gray-800:      #C8C2B8;
  --color-gray-600:      #8C877F;
  --color-gray-400:      #5A5550;
  --color-gray-200:      #2E2B27;
  --text-display:        #0A0A0A;
  --text-primary:        #1A1815;
  --text-secondary:      #6B6560;
  --text-disabled:       #B0AA9F;
  --color-accent:        #D71921;
  --color-success:       #16A34A;
  --color-warning:       #D97706;
  --canvas-dot-color:    rgba(0, 0, 0, 0.08);
  --canvas-grid-color:   rgba(0, 0, 0, 0.04);
  --canvas-bg:           #F5F2EE;
  --glass-tint:          rgba(255, 255, 255, 0.55);
  --glass-specular:      rgba(255, 255, 255, 0.70);
  --glass-bezel-shadow:  rgba(0, 0, 0, 0.08);
  --glass-cast-shadow:   rgba(0, 0, 0, 0.15);
  --glass-frost:         rgba(240, 236, 229, 0.75);
  --state-hover:         rgba(0, 0, 0, 0.05);
  --state-active:        rgba(0, 0, 0, 0.10);
  --state-focus-ring:    2px solid rgba(0, 0, 0, 0.50);
  --state-focus-offset:  3px;
  --state-disabled-opacity: 0.35;
}

/* ── REDUCED MOTION ──────────────────────────── */
@media (prefers-reduced-motion: reduce) {
  *, *::before, *::after {
    animation-duration:        0.01ms !important;
    animation-iteration-count: 1      !important;
    transition-duration:       0.01ms !important;
    scroll-behavior:           auto   !important;
  }
}
```

PART XVI — MODE SWITCH IMPLEMENTATION

```
// src/utils/theme.js
```

```
const STORAGE_KEY = 'lg-theme'

export const getInitialTheme = () => {
  const saved = localStorage.getItem(STORAGE_KEY)
  if (saved === 'dark' || saved === 'light') return saved
  return window.matchMedia('(prefers-color-scheme: dark)').matches ? 'dark' : 'li
}

export const applyTheme = (theme) => {
  document.documentElement.setAttribute('data-theme', theme)
  localStorage.setItem(STORAGE_KEY, theme)
  document.querySelector('meta[name="theme-color"]')
    ?.setAttribute('content', theme === 'dark' ? '#0A0A0A' : '#F5F2EE')
  window.dispatchEvent(new CustomEvent('theme-change', { detail: { theme } }))
}

export const toggleTheme = () => {
  const current = document.documentElement.getAttribute('data-theme') || 'dark'
  applyTheme(current === 'dark' ? 'light' : 'dark')
}
```

Apply theme before first paint to prevent flash:

```
<script>
  const saved = localStorage.getItem('lg-theme')
  const theme = saved || (window.matchMedia('(prefers-color-scheme: dark)').match
  document.documentElement.setAttribute('data-theme', theme)
</script>
```

PART XVII — IMPLEMENTATION BUILD ORDER

Execute in sequence. Each phase must be verified before proceeding.

Phase 1 — Token Foundation

```
Files:
  src/styles/tokens.css          ← Part XV above (copy verbatim)
  src/styles/reset.css           ← CSS reset + base elements
  src/styles/canvas.css          ← Canvas world + dot grid
  src/utils/theme.js             ← Part XVI above (copy verbatim)
  index.html                     ← Add theme init script before </head>

Verify:
```

```
  [ ] Dark mode renders on OLED device — no bleed at #0A0A0A
  [ ] Light mode warm paper feel — not clinical white
  [ ] Font trio (Space Grotesk, Space Mono, Doto) loads before paint
  [ ] Mode toggle persists across page refresh
  [ ] OS preference detected on first launch
```

Phase 2 — Glass Renderer

```
Files:
  src/webgpu/GlassRenderer.ts    ← jeantimex pipeline wrapper
  src/webgpu/uniforms.ts         ← All 10 surface param objects (5 × dark + light
  src/webgpu/fallback.css        ← Tier 2 + Tier 3 CSS glass (from Part IV)
  src/utils/glassTier.ts         ← WebGPU / CSS / flat detection

Verify:
  [ ] Tier 1 (WebGPU) renders on Chrome 113+
  [ ] Tier 2 (backdrop-filter) renders on Safari
  [ ] Tier 3 (flat) renders on Firefox without flag
  [ ] theme-change event updates all surface uniforms
  [ ] Reduced motion disables liquidDeformation on all surfaces
```

Phase 3 — Canvas Engine

```
Files:
  src/canvas/CanvasEngine.ts     ← Pan, zoom, viewport, virtual culling
  src/canvas/DragManager.ts      ← Drag, snap-to-grid, alignment guides
  src/canvas/HistoryManager.ts   ← Undo/redo command pattern (100 op stack)
  src/canvas/SelectionManager.ts ← Multi-select, group operations

Verify:
  [ ] 60fps pan/zoom at 100 cards — Chrome DevTools Performance
  [ ] 55fps+ at 500 cards
  [ ] Drag positions persist across page reload (IndexedDB)
  [ ] Alignment guides appear at ±8px from other card edges
  [ ] Undo/redo works for: move, create, delete, resize, group
```

Phase 4 — Card Components

```
Files:
  src/cards/BaseCard.tsx         ← Shared: glass, drag, states, mode-awareness
  src/cards/BookmarkCard.tsx     ← Twitter-sourced card
  src/cards/WebClipCard.tsx      ← URL paste card
  src/cards/NoteCard.tsx         ← Tiptap editor card
```

```
  src/cards/ImageCard.tsx       ← Upload/paste image
  src/cards/GroupCard.tsx       ← Stack container
  src/cards/cards.css           ← Shared card styles

Verify:
  [ ] All five states render correctly on all card types
  [ ] Glass uniforms update on mode switch — no stale dark glass in light mode
  [ ] Drag lift animation fires with correct spring values
  [ ] Empty card state shows [LOADING...] during metadata fetch
  [ ] Card accent dot renders above glass layer (never refracted)
```

Phase 5 — UI Chrome

```
Files:
  src/ui/Toolbar.tsx            ← Desktop left rail + mobile bottom bar
  src/ui/CommandPalette.tsx     ← ⌘K overlay
  src/ui/ContextMenu.tsx        ← Right-click (desktop) / long-press (mobile)
  src/ui/BottomSheet.tsx        ← Mobile bottom sheet, three snap points
  src/ui/Minimap.tsx            ← Canvas overview, 30fps update
  src/ui/SpacesSidebar.tsx      ← Spaces navigation panel
  src/ui/Lightbox.tsx           ← Full-screen media viewer
  src/ui/ModeToggle.tsx         ← Mechanical sliding toggle

Verify:
  [ ] All glass surfaces at correct tier
  [ ] Full keyboard navigation (Tab, arrows, Esc) through all chrome
  [ ] Bottom sheet three snap points with correct spring values
  [ ] Mode toggle animates correctly in both directions
  [ ] Minimap viewport rect tracks canvas pan/zoom in real time
```

Phase 6 — Icons & Typography Audit

```
  [ ] pnpm add @phosphor-icons/react
  [ ] All icons from vocabulary list only — remove any others
  [ ] Max 3 font sizes active simultaneously per screen
  [ ] All toolbar/metadata labels: Space Mono ALL CAPS
  [ ] All body/card content: Space Grotesk
  [ ] Doto used for exactly one hero element per canvas view
  [ ] text-shadow on all body text above glass surfaces
```

Phase 7 — Accessibility

```
  [ ] All ARIA roles from Part XIV applied
```

```
[ ] canvas-announcer and canvas-error live regions present
[ ] Keyboard-only navigation end-to-end (no mouse required)
[ ] pnpm add -D @axe-core/react — 0 critical violations
[ ] VoiceOver (macOS) test — canvas readable
[ ] Reduced-motion: WebGPU deformation disabled, CSS transitions 0.01ms
```

Phase 8 — Responsive & PWA

```
[ ] Three layout modes tested: mobile / tablet / desktop
[ ] manifest.json: start_url="/looking-glass/", scope="/looking-glass/"
[ ] Service worker caches app shell
[ ] Install to iOS Safari home screen — launch without errors
[ ] Safe-area-inset applied to toolbar and bottom sheet
[ ] All touch gestures from Part XII functional
[ ] Lighthouse PWA score > 90
```

Phase 9 — Performance

```
[ ] First Contentful Paint < 1.2s (Lighthouse)
[ ] Time to Interactive < 2.0s (Lighthouse)
[ ] Canvas pan 100 cards: 60fps
[ ] Canvas pan 500 cards: 55fps+
[ ] Initial bundle gzipped < 150KB (Vite Bundle Analyzer)
[ ] WebGPU glass init < 200ms
[ ] Mode switch: no dropped frames
[ ] IndexedDB write (single item) < 10ms
```

PART XVIII — WHAT THIS SHOULD FEEL LIKE

Someone opening Looking Glass on their iPhone for the first time experiences two things in sequence.

First: "This looks like a serious tool." The Nothing typography — sharp, monochrome, three-layer hierarchy — signals precision. It is not a consumer app covered in gradients. It is an instrument.

Second: "Did that just move?" When they drag their first card, the spring deformation of the glass surface is unexpected. Hardware quality in a browser. The card lifts, tilts slightly toward the direction of drag, then when dropped — settles with physical weight.

On dark mode the app is nocturnal and instrument-like. Glass surfaces glow at their edges.

Cards float in dark space.

On light mode the app is like a beautifully printed document. Glass surfaces cast shadows on warm paper. Cards feel weighted.

The two modes are not themes applied to the same design. They are the same design expressed through two different materials. Both are correct. Both are complete.

Sources: `dominikmartn/nothing-design-skill` SKILL.md v3.0.0 · `jeantimex/glass-effect-webgpu`